

2(a)	<i>any one from:</i> 1 <i>idea that</i> (cold buffer) prevents, (mitochondrial) enzymes / proteins , denaturing ; 2 <i>idea that</i> (cold temperature) prevents damage to mitochondria, by enzymes ; 3 <i>idea that</i> (buffer) prevents (osmotic) lysis of mitochondria ;	1
2(b)(i)	49.50 ;	1

Question	Answer	Marks																																				
2(b)(ii)	correct identification of anomalous results ;; <table><tr><td></td><td colspan="5">rate of oxygen consumption</td></tr><tr><td></td><td>trial 1</td><td>trial 2</td><td>trail 3</td><td>trial 4</td><td>mean</td></tr><tr><td>mitochondria alone</td><td>0.03</td><td>0.02</td><td>0.03</td><td>0.01</td><td>0.02</td></tr><tr><td>succinate added</td><td>50.22</td><td>49.10</td><td>48.53</td><td>50.15</td><td>49.50</td></tr><tr><td>ADP and succinate added</td><td>139.23</td><td>170.10</td><td>142.67</td><td>138.10</td><td>147.53</td></tr><tr><td>ADP, succinate and cyanide added</td><td>0.00</td><td>0.10</td><td>0.00</td><td>0.01</td><td>0.03</td></tr></table>		rate of oxygen consumption						trial 1	trial 2	trail 3	trial 4	mean	mitochondria alone	0.03	0.02	0.03	0.01	0.02	succinate added	50.22	49.10	48.53	50.15	49.50	ADP and succinate added	139.23	170.10	142.67	138.10	147.53	ADP, succinate and cyanide added	0.00	0.10	0.00	0.01	0.03	2
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2(b)(iii)	1 do not include, anomalies and calculate (a more accurate) mean ; 2 calculate, standard deviation / standard error / 95% confidence limits (for each trial) ;	2																																				

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2(c)	any three from: 1 in mitochondria (alone): oxygen concentration does not change or do not use oxygen or rate of oxygen consumption is very low ; 2 adding succinate (and no ADP) increases the rate of, <u>aerobic</u> respiration / Krebs cycle / oxidative phosphorylation ; 3 adding succinate and ADP, gives a larger increase in the rate of, <u>aerobic</u> respiration / Krebs cycle / oxidative phosphorylation ; 4 oxygen acts as the final electron acceptor (in aerobic respiration / oxidative phosphorylation) ; 5 adding cyanide stops, (aerobic) respiration / oxidative phosphorylation ;	3